Project Brief — P4A 2026

Overview

The group project runs across Sessions 3 to 5. You will load, explore, and visualise a real job postings dataset, build a reusable Python module, and use a local language model to interpret your findings. You submit your work after Session 5.

Research question: Which industries are most active on LinkedIn, what benefits do they typically offer, and which postings attract the most views?

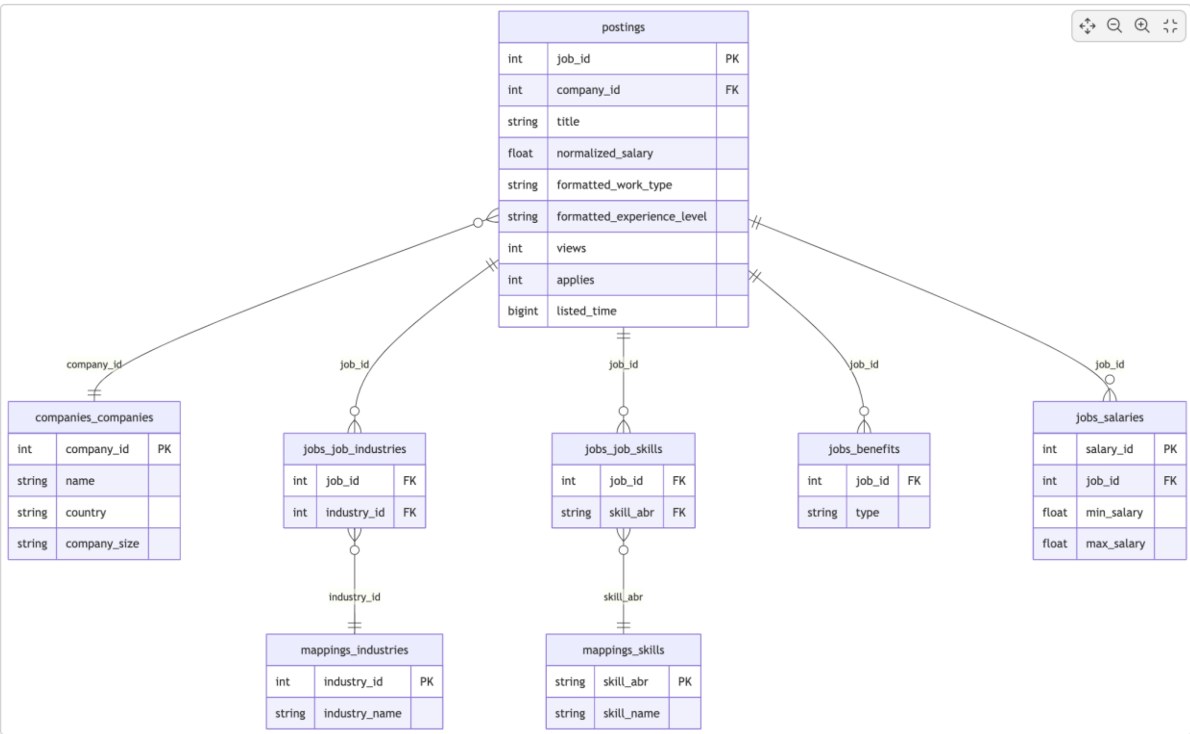
Task sheets are distributed at the end of each session and guide the work for that phase. This document sets out the rules, required deliverables, and submission format.

Dataset

LinkedIn Job Postings (2023–2024) Source: Kaggle — arshkon/linkedin-job-postings Licence: CC BY-SA 4.0 Author: Arsh K

The dataset contains 40,059 job postings scraped from LinkedIn, stored across multiple related tables. You access it through the shared course database — no download required.

Database schema



Notation	Meaning
||	exactly one
o|	zero or one
||--o{	one-to-many
}o--||	many-to-one

Example: postings }o--\|\| companies_companies — each posting belongs to exactly one company; a company may have zero or many postings.

postings_with_benefits: as part of the SQL phase, your group will create a staging table called postings_with_benefits in your database. This table combines the postings table with the jobs_benefits table and filters to active postings only. Your Python code loads from this table — it must exist before you run load_postings().

Attribution requirement: Your slides must include the following line on the dataset slide or in the footer:

Dataset: *LinkedIn Job Postings 2023–2024*, Arsh K, Kaggle. Licence: CC BY-SA 4.0.

Groups

Groups of 3–4 students. Groups are fixed for the full project. Work is submitted once per group.

Deliverables

1. Code — submitted as .zip via Virtual Campus

Required folder structure:

```
project_groupname/
├── analysis.py
├── notebook.ipynb
├── output/
└── sql/
    └── staging.sql
```

analysis.py must contain at least these four functions, each with a docstring:

```
def load_postings(engine) -> pd.DataFrame:
    """Load the postings_with_benefits staging table from the course database."""

def filter_fulltime(df: pd.DataFrame) -> pd.DataFrame:
    """Filter to full-time postings. Returns a copy."""
```

```python
def company_summary(engine) -> pd.DataFrame:
    """Load the companies_companies table, filter to US companies."""

def top_industries(df_postings, df_job_industries, df_industries, n: int) -> pd.DataFrame:
    """Return the top N industries by posting count."""
```

notebook.ipynb must:

- Import and call all four functions from analysis.py
- Run from top to bottom without errors, with all outputs visible
- Include at least one chart with a title and labelled axes
- Use markdown cells to explain each section in plain language

sql/01_setup.sql — creates the tables and imports the data; each section commented
sql/02_queries.sql — at least 3 analytical queries; each query preceded by a comment explaining what it answers

Data files are not submitted. All groups use the shared course database.

2. Slides — submitted as PDF via Virtual Campus

8–15 slides, submitted as PDF. Appendix slides do not count toward the limit. Each slide below corresponds to work you will complete across the project sessions.

Slide	Content
Title	Group name, class, research question
Dataset overview	What the dataset contains and why postings_with_benefits is the starting point
SQL findings (×2)	Results of at least 2 analytical queries — what do they show?
Company landscape	US share, size distribution, top countries — one insight sentence
Benefits landscape	Top benefits chart — what do companies typically offer on LinkedIn?
Industry analysis	Top 15 industries chart — which sectors dominate?
Benefits × industry	Which industries offer medical insurance most relative to posting volume?
Anomaly detection	Outlier counts (Z-score vs IQR), which method you chose and why
LLM prompt	The prompt you sent and the data you passed to the model
LLM response + critique	Excerpt of the response and your one-sentence critical assessment

Key insights	3–5 bullet points: what does this dataset tell you about the LinkedIn job market?
Attribution	Dataset credit (required — see attribution requirement above)

Submission Deadlines

Class	Deadline
Class 2_2	26 May 2026, 23:59

Both deliverables must be submitted via Virtual Campus before the deadline. Late submissions are not accepted.

Rules

1. **All group members must understand the full submission.** Any member may be asked to explain a code decision or make a small live modification. This is part of the assessment.
2. **The notebook must show your process.** Keep draft cells, commented-out attempts, and visible iteration.
3. **AI tools are permitted, but understanding is required.** If you used an AI tool to generate code, you are responsible for understanding it and adapting it to the data. You will be asked about it.
4. **The dataset attribution must appear in your slides.** Submissions without it are incomplete.
5. **Do not submit data files or generated outputs.** Everyone uses the shared course database. Include only code, the notebook, and SQL files.

Assessment

Component	Weight	Notes
Project	40%	Code + slides, submitted after Session 5

Extraordinary evaluation - Project grade cannot be recovered.